

A proof of Erdős Problem 689

Working note

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Abstract

We prove that for all sufficiently large n one can choose one residue class $a_p \pmod{p}$ for each prime $p \leq n$ so that every integer $m \in [1, n]$ lies in at least two of the chosen classes. The proof starts with the parity-first assignment $a_2 \equiv 1 \pmod{2}$, leaves 3 in the zero class, switches a fixed large set of small primes $S \subset \{7, 11, 13, \dots\}$ to nonzero classes, and then uses robust cleanup primes $P > n/5$. The cleanup is reduced to a finite-core prime-difference matching problem. A half-residue transport kernel produces a fractional matching with fixed slack; the finite-complexity Green–Tao–Ziegler theorem for affine-linear forms in primes gives the required first and second weighted moments; and Kahn’s fractional Frankl–Rödl–Pippenger theorem rounds the fractional matching to an integral matching. The exact form of Kahn’s theorem used here is Theorem 1.5 of [4].

1 The parity-first residual problem

For a prime $p \leq n$, write $a_p \pmod{p}$ for the residue class chosen for p . We begin with

$$a_2 \equiv 1 \pmod{2},$$

leave all odd primes in the zero class, and later switch selected odd primes to nonzero classes. The baseline coverage is

$$C_0(m) = \mathbf{1}_{m \text{ odd}} + \#\{p \text{ odd prime} : p \mid m\}.$$

If R is the set of switched odd primes, b_p is the new residue for $p \in R$,

$$L_R(m) = \#\{p \in R : p \mid m\}, \quad G_R(m) = \#\{p \in R : m \equiv b_p \pmod{p}\},$$

then the exact requirement is

$$G_R(m) \geq \max(0, 2 - C_0(m) + L_R(m)). \tag{1}$$

Thus switching a prime creates debt only at multiples whose previous coverage used that prime’s zero class.

We keep 3 at 0 $\pmod{3}$. Choose a fixed finite set

$$S \subset \{7, 11, 13, \dots\}$$

and switch each $s \in S$ to a nonzero residue $b_s \pmod{s}$. Put

$$H_S(x) = \#\{s \in S : x \equiv b_s \pmod{s}\}.$$

After these fixed switches, the main residual tokens are even integers

$$x = 2^k u q, \quad k \geq 1, \quad (2)$$

where u is odd and S -smooth, $q \notin S$ is an odd prime occurring to exponent one, and $H_S(x) = 0$. Prime-power exceptions, pure $\{2\} \cup S$ -smooth terms, $m = 1$, and other boundary cases contribute $o(n/\log n)$ residual tokens.

Let $A_S(n)$ denote the main residual token set.

Lemma 1 (Residual asymptotic and exceptional tokens). *For fixed S ,*

$$|A_S(n)| = (1 + o(1)) \frac{n}{\log n}. \quad (3)$$

Moreover the number of residual tokens not included in $A_S(n)$ is

$$O_S\left(\sqrt{n} + (\log n)^{|S|+1}\right) = o(n/\log n).$$

Finally, if a finite coefficient core omits coefficient weight at most ε in the sense of the proof below, then the number of omitted main residual targets is $O_S(\varepsilon n/\log n) + o(n/\log n)$.

Proof. Let $\mathcal{D}_S$ be the odd S -smooth numbers. For $d \in \mathcal{D}_S$ put

$$\Theta_S(d) = \prod_{\substack{s \in S \\ s \nmid d}} \frac{s-2}{s-1}.$$

For fixed d and $k \geq 1$, the residuals $2^k d q \leq n$ with $q \notin S$ prime are counted in a fixed union of reduced residue classes modulo W , hence fixed-modulus PNT gives

$$\#\{q : 2^k d q \leq n, H_S(2^k d q) = 0\} = \Theta_S(d) \frac{n}{2^k d \log n} + o_{d,k}(n/\log n). \quad (3a)$$

The coefficient sum is

$$\sum_{d \in \mathcal{D}_S} \frac{\Theta_S(d)}{d} = \prod_{s \in S} \left(\frac{s-2}{s-1} + \sum_{e \geq 1} \frac{1}{s^e} \right) = 1, \quad \sum_{k \geq 1} 2^{-k} = 1. \quad (3b)$$

To justify summing the asymptotics, choose a finite set $\mathcal{K} \subset \mathcal{D}_S \times \mathbb{N}$ so that

$$\sum_{(d,k) \notin \mathcal{K}} \frac{\Theta_S(d)}{2^k d} < \varepsilon.$$

On the finite set $\mathcal{K}$, (3a) may be summed directly. On the tail with $2^k d \leq n^{1/2}$, Brun–Titchmarsh or the standard fixed-modulus upper bound $\pi(x; W, a) \ll_S x/\log x$ gives $O_S(\varepsilon n/\log n)$. On the tail with $2^k d > n^{1/2}$, the trivial bound by $\pi(n/(2^k d)) \leq n^{1/2}$ and the estimate

$$\#\{(d, k) : d \in \mathcal{D}_S, 2^k d \leq n\} \ll_S (\log n)^{|S|+1}$$

give $o(n/\log n)$. Letting $\varepsilon \rightarrow 0$ proves (3).

It remains to bound exceptional tokens. Odd S -smooth integers and even $\{2\} \cup S$ -smooth integers contribute $O_S((\log n)^{|S|+1})$ tokens. Integers of the form $2^k dq^a \leq n$ with $a \geq 2$, $d \in \mathcal{D}_S$, and $q \notin S$ prime contribute at most

$$\sum_{d,k} \#\{q^a \leq n/(2^k d) : a \geq 2\} \ll_S \sqrt{n} \sum_{d \in \mathcal{D}_S} \frac{1}{\sqrt{d}} \sum_{k \geq 1} 2^{-k/2} \ll_S \sqrt{n}.$$

This proves the exceptional-token bound. The finite-core tail estimate is the same tail argument with $\mathcal{K}$ chosen to be the coefficient core. $\square$

Define

$$A_1(n) = \{x \in A_S(n) : v_2(x) = 1\}, \quad A_2(n) = \{x \in A_S(n) : v_2(x) \geq 2\}.$$

Then

$$|A_1(n)| = \left(\frac{1}{2} + o(1)\right) |A_S(n)|, \quad |A_2(n)| = \left(\frac{1}{2} + o(1)\right) |A_S(n)|. \quad (4)$$

2 Robust cleanup primes

Let

$$W = \prod_{s \in S} s.$$

For a unit class $\pi \pmod{W}$, call π robust if

$$H_S(\pi) \geq 1, \quad H_S(2\pi) \geq 2, \quad H_S(4\pi) \geq 2. \quad (5)$$

Let $\mathcal{B} \subset (\mathbb{Z}/W\mathbb{Z})^\times$ be the robust set and set

$$\delta_S = \frac{|\mathcal{B}|}{\varphi(W)}.$$

For $s \in S$, the three local classes

$$\pi \equiv b_s, \quad 2\pi \equiv b_s, \quad 4\pi \equiv b_s \pmod{s}$$

are distinct because $s \geq 7$. Hence δ_S depends only on S , not on the particular nonzero residues b_s .

Lemma 2 (Robust density). *There are fixed finite sets $S \subset \{7, 11, 13, \dots\}$ for which δ_S is arbitrarily close to 1.*

Proof. Let

$$A_S^{(0)} = \prod_{s \in S} \left(1 - \frac{1}{s-1}\right), \quad B_S = \sum_{s \in S} \frac{1}{s-2}.$$

If X_1, X_2, X_4 count hits to $\pi, 2\pi, 4\pi$, then

$$\delta_S = \mathbb{P}(X_1 \geq 1, X_2 \geq 2, X_4 \geq 2).$$

Now

$$\mathbb{P}(X_1 = 0) = A_S^{(0)},$$

and

$$\mathbb{P}(X_2 < 2) = A_S^{(0)}(1 + B_S),$$

with the same estimate for X_4 . Thus

$$\delta_S \geq 1 - A_S^{(0)}(3 + 2B_S). \quad (6)$$

Taking $S(y) = \{p : 7 \leq p \leq y, p \text{ prime}\}$, Mertens' theorem and the prime number theorem give $A_{S(y)}^{(0)} \asymp 1/\log y$ and $B_{S(y)} = O(\log \log y)$, so the right side of (6) tends to 1. $\square$

Set

$$\beta_* = \frac{1}{2} \left(1 - \frac{3}{5} e^{-2} \right) = 0.459399 \dots,$$

$$\delta_* = \frac{1}{\beta_* + 3/5} = 0.943931 \dots$$

Choose S with $\delta_S > \delta_*$, and choose

$$\delta_S^{-1} - \frac{3}{5} < \beta < \beta_*. \quad (7)$$

Let

$$\Delta = (\beta + 3/5)\delta_S - 1 > 0. \quad (8)$$

This fixed positive margin absorbs coefficient tails and exceptional tokens. Define

$$\mathcal{R}_\beta(n) = \{P \in (n/5, \beta n] : P \text{ prime}, P \bmod W \in \mathcal{B}\}$$

and

$$\mathcal{R}(n) = \{P \in (n/5, n] : P \text{ prime}, P \bmod W \in \mathcal{B}\}.$$

Fixed-modulus PNT gives

$$|\mathcal{R}_\beta(n)| = ((\beta - 1/5)\delta_S + o(1)) \frac{n}{\log n}, \quad (9)$$

$$|\mathcal{R}(n)| = \left(\frac{4}{5} \delta_S + o(1) \right) \frac{n}{\log n}. \quad (10)$$

Lemma 3 (Robust side debt). *Switching any robust prime $P > n/5$ to any nonzero residue creates no unresolved side debt.*

Proof. The only multiples of P not exceeding n are $P, 2P, 3P, 4P$. The integer P is odd and has the parity hit plus at least one S -hit. The integers $2P$ and $4P$ have at least two S -hits by (5). The integer $3P$ has the parity hit and the unchanged zero class modulo 3. There is no multiple $5P \leq n$. $\square$

3 The finite-core hypergraph and the half-residue kernel

Choose finite coefficient cores

$$\mathcal{A}_1 \subset \{a : a \text{ odd and } S\text{-smooth}\},$$

$$\mathcal{A}_2 \subset \{b : b = 2^j u, j \geq 1, u \text{ odd and } S\text{-smooth}\}.$$

Let $X_n \subset A_1(n)$ be the residual targets $x = 2aq$ with $a \in \mathcal{A}_1$, and let $Y_n \subset A_2(n)$ be the residual targets $y = 2bq'$ with $b \in \mathcal{A}_2$. The cores will be chosen so that they capture fractions α_X, α_Y of the full coefficient mass, uniformly in the half-residue classes below, and

$$\alpha_X, \alpha_Y > G(\beta) + \eta \quad (11)$$

for some fixed $\eta > 0$, where

$$G(\beta) = \int_{1/5}^{\beta} \frac{dt}{1-2t} = \frac{1}{2} \log \left(\frac{3/5}{1-2\beta} \right) < 1. \quad (12)$$

This is possible because $\beta < \beta_*$. In the explicit notation below this means

$$\alpha_X = \min_{A \in \mathcal{C}} 2\Xi_A, \quad \alpha_Y = \min_{A \in \mathcal{C}} 2H_A.$$

The quantifiers are chosen in this order: first choose S, δ_S, β with strict margin in (7), then choose $\eta > 0$ with $G(\beta) + \eta < 1$, and finally choose the finite cores so that $\alpha_X, \alpha_Y > G(\beta) + \eta$ and the omitted coefficient-tail mass is smaller than the cleanup surplus.

Let

$$c_s \equiv 2^{-1}b_s \pmod{s},$$

and define

$$\mathcal{C} = \{A \pmod{W} : A \not\equiv c_s \pmod{s} \text{ for every } s \in S\}.$$

For $x = 2aq$ write $A \equiv aq \pmod{W}$, and for $y = 2bq'$ write $B \equiv bq' \pmod{W}$. Then residual membership is $A, B \in \mathcal{C}$.

Lemma 4 (Half-residue regularity). *For every unit $\pi \pmod{W}$ and every $\sigma \in \{\pm 1\}$,*

$$\#\{A \in \mathcal{C} : A + \sigma\pi \in \mathcal{C}\} = M := \prod_{s \in S} (s - 2). \quad (13)$$

Moreover, the full infinite S -smooth coefficient distribution is uniform on $\mathcal{C}$ on both the X - and Y -sides, and finite cores can approximate this uniformly.

Proof. For (13), modulo each $s \in S$ the conditions are $A \neq c_s$ and $A \neq c_s - \sigma\pi$. Since π is a unit, these are two distinct forbidden classes, leaving $s - 2$ choices.

For the coefficient distribution, fix $s \in S$. If the s -adic exponent of the coefficient is 0, then for every nonzero half-residue $A \pmod{s}$ there is one reduced class of the prime variable, giving local mass $1/(s - 1)$. If the exponent is at least 1, then $A \equiv 0 \pmod{s}$ and the coefficient weight is $\sum_{e \geq 1} s^{-e} = 1/(s - 1)$, with the prime variable unrestricted among reduced classes. Thus every allowed local half-residue has the same mass. Multiplying over $s \in S$ gives uniformity on $\mathcal{C}$. Truncating exponents to $e \leq E$ changes only the $A \equiv 0 \pmod{s}$ local mass by $O(s^{-E})$, so since S is fixed, large finite cores approximate the uniform distribution uniformly. The extra 2-power in the Y -coefficients contributes a common factor $\sum_{1 \leq j \leq J} 2^{-j}$ and does not affect half-residue uniformity. $\square$

Define the 3-partite 3-uniform hypergraph

$$H_n \subset X_n \times Y_n \times Z_n, \quad Z_n = \mathcal{R}_\beta(n),$$

by putting an edge (x, y, P) when

$$|y - x| = 2P. \quad (14)$$

The parity split is essential: X_n consists of targets $v_2 = 1$, while Y_n consists of targets $v_2 \geq 2$, so $(y - x)/2$ is odd.

The aggregate continuum transport is as follows. A label is (t, π) with $t \in (1/5, \beta]$ and $\pi \in \mathcal{B}$. Choose $\sigma = \pm 1$ with probability $1/2$, choose uniformly an $A \in \mathcal{C}$ with $A + \sigma\pi \in \mathcal{C}$, and choose $u \in (0, 1 - 2t)$ uniformly. If $\sigma = +1$, send mass to

$$X = (u, A), \quad Y = (u + 2t, A + \pi),$$

and if $\sigma = -1$, send mass to

$$X = (u + 2t, A), \quad Y = (u, A - \pi).$$

This gives exact label load 1. The side load at a half-residue class is at most

$$\int_{1/5}^{\beta} \frac{dt}{1 - 2t} = G(\beta).$$

For finite cores this is multiplied by α_X^{-1} or α_Y^{-1} , hence by (11) the limiting side loads are at most $1 - 2\gamma$ for some fixed $\gamma > 0$.

The aggregate transport lifts to typed kernels. A type is

$$\tau = (a, b, \sigma, r, r', \pi),$$

where $a \in \mathcal{A}_1$, $b \in \mathcal{A}_2$,

$$\begin{aligned} q &\equiv r \pmod{W}, & q' &\equiv r' \pmod{W}, \\ \pi &\equiv \sigma(br' - ar) \pmod{W}, & \pi &\in \mathcal{B}. \end{aligned}$$

The support polygon is

$$D_\tau = \left\{ (Q, Q') : 0 < Q \leq \frac{1}{2a}, 0 < Q' \leq \frac{1}{2b}, \frac{1}{5} < \sigma(bQ' - aQ) \leq \beta \right\}. \quad (15)$$

Every residue-compatible type is admissible. Indeed, modulo 2 the parity split gives $P = \sigma(bq' - aq)$ odd; modulo every $s \in S$, the prescribed classes have q, q' and π units; and outside $2W$ the three forms $q, q', \sigma(bq' - aq)$ are three distinct lines, so their zero sets cannot cover all residue pairs. Let $\lambda_\tau > 0$ be the fixed edge-total constant in the Green–Tao–Ziegler asymptotic. The typed kernel g_τ is the lifted aggregate density divided by λ_τ . It is bounded because the coefficient core is finite, $\beta < 1/2$, and every λ_τ is fixed and positive. The condition $\gcd(a, b) = 1$ is automatic: if some $s \in S$ divided both a and b , then $A \equiv B \equiv 0 \pmod{s}$, but admissibility gives $B - A \equiv \sigma\pi \not\equiv 0 \pmod{s}$.

We record the typed lift explicitly. For a half-residue $A \in \mathcal{C}$ set

$$\Xi_A = \sum_{\substack{a \in \mathcal{A}_1, r \\ ar \equiv A \pmod{W}}} \frac{1}{2a}, \quad H_A = \sum_{\substack{b \in \mathcal{A}_2, r' \\ br' \equiv A \pmod{W}}} \frac{1}{2b}.$$

The finite cores are chosen so that

$$\frac{1 - \varepsilon}{2} \leq \Xi_A, H_A \leq \frac{1}{2} \quad (A \in \mathcal{C}), \quad (15a)$$

with $\varepsilon > 0$ small enough that $1 - \varepsilon > G(\beta) + \eta/2$. Equivalently, $\alpha_X, \alpha_Y \geq 1 - \varepsilon$, so this is the uniform version of (11). Define conditional weights

$$\rho_X^A(a, r) = \frac{1/(2a)}{\Xi_A}, \quad \rho_Y^B(b, r') = \frac{1/(2b)}{H_B}.$$

For $t \in (1/5, \beta]$ and $A + \sigma\pi \in \mathcal{C}$, let

$$K_\sigma(t, \pi; u, A) = \frac{1}{2M(1-2t)} \quad (0 < u < 1-2t),$$

and let it be zero otherwise. If $(Q, Q') \in D_\tau$, put

$$t = \sigma(bQ' - aQ), \quad u_\tau(Q, Q') = \begin{cases} 2aQ, & \sigma = +1, \\ 2bQ', & \sigma = -1. \end{cases}$$

The change of variables $(Q, Q') \mapsto (u_\tau, t)$ has Jacobian $2ab$. With the GTZ normalization below, the base edge measure of type τ is $\lambda_\tau dQ dQ'$. The typed kernel is therefore

$$g_\tau(Q, Q') = \frac{2ab}{\lambda_\tau} K_\sigma(t, \pi; u_\tau(Q, Q'), A) \rho_X^A(a, r) \rho_Y^B(b, r'). \quad (15b)$$

This formula gives the load identities. For labels, the coarea factor is $b^{-1}dQ$, and after summing the conditional distributions over all typed fibres above a fixed half-residue pair $(A, A + \sigma\pi)$ one obtains

$$L_Z^{\lim}(\pi, t) = \sum_{\sigma=\pm 1} \sum_{\substack{A \in \mathcal{C} \\ A + \sigma\pi \in \mathcal{C}}} \int_0^{1-2t} K_\sigma(t, \pi; u, A) du = 1. \quad (15c)$$

For the X -side, writing $z = 2aQ$, the same change of variables gives

$$L_X^{\lim}(a, r, Q) = \frac{1}{\Xi_A} \left[\frac{N_+(A)}{2M} J_\beta\left(\frac{1-z}{2}\right) + \frac{N_-(A)}{2M} J_\beta\left(\frac{z}{2}\right) \right], \quad (15d)$$

where

$$N_\pm(A) = \#\{\pi \in \mathcal{B} : A \pm \pi \in \mathcal{C}\}, \quad J_\beta(u) = \int_{1/5}^{\min(\beta, u)} \frac{dt}{1-2t}$$

with the integral interpreted as 0 if $u \leq 1/5$. Since $N_\pm(A) \leq M$ and (15a) holds,

$$L_X^{\lim} \leq \frac{1}{1-\varepsilon} \left[J_\beta\left(\frac{1-z}{2}\right) + J_\beta\left(\frac{z}{2}\right) \right] \leq \frac{G(\beta)}{1-\varepsilon} \leq 1-2\gamma.$$

The middle inequality uses $\beta > 2/5$, which follows from (7) and $\delta_S \leq 1$: if $u + v = 1/2$, then $J_\beta(u) + J_\beta(v) \leq G(\beta)$. The Y -side is identical with Ξ_A replaced by H_B . This proves the finite-core typed-kernel lift and the side slack used below.

4 GTZ weighted moments

For an edge $e = (x, y, P)$ of type τ , write

$$x = 2aq, \quad y = 2bq', \quad P = \sigma(bq' - aq),$$

and assign

$$w_e = \frac{\log^2 n}{n} g_\tau(q/n, q'/n). \quad (16)$$

The following proposition is the only analytic input.

Proposition 5 (Weighted GTZ moment proposition). *The loads*

$$L_Z(P) = \sum_{e \ni P} w_e, \quad L_X(x) = \sum_{e \ni x} w_e, \quad L_Y(y) = \sum_{e \ni y} w_e$$

satisfy

$$\sum_{P \in Z_n} (L_Z(P) - 1)^2 = o(|Z_n|), \quad (17)$$

$$\sum_{x \in X_n} (L_X(x) - L_X^{\text{lim}}(x))^2 = o(|X_n|), \quad (18)$$

and

$$\sum_{y \in Y_n} (L_Y(y) - L_Y^{\text{lim}}(y))^2 = o(|Y_n|), \quad (19)$$

where $L_X^{\text{lim}}, L_Y^{\text{lim}} \leq 1 - 2\gamma$.

Proof. We use the finite-complexity Green–Tao theorem for affine-linear forms in primes, made unconditional by the Green–Tao nilsequence theorem and the Green–Tao–Ziegler inverse theorem [1, 2, 3]. It is cleanest here to use the normalized auxiliary W -tricked form, because then the main terms are normalized residue cell averages and no singular-series cancellation has to be checked separately.

Concretely, put $W_0 = 2W$. For an auxiliary parameter w let

$$W_{\text{aux}} = W_0 \prod_{\substack{\ell \leq w \\ \ell \nmid W_0}} \ell.$$

All congruence conditions modulo the fixed W_0 are kept as residue data, and one then sums over residue lifts modulo W_{aux} on which all displayed linear forms are units. For each fixed w , the GTZ theorem gives the expected normalized average for every fixed finite-complexity system on every rational polytope. After taking $n \rightarrow \infty$, one lets $w \rightarrow \infty$. Equivalently, in raw prime-indicator notation this says that, for a fixed finite-complexity system $\Psi = (\psi_1, \dots, \psi_t)$ on a fixed affine lattice and a bounded Riemann-integrable weight F ,

$$\sum_{\mathbf{m} \in nK \cap \Lambda} F(\mathbf{m}/n) \prod_{i=1}^t \mathbf{1}_{\psi_i(\mathbf{m}) \in \mathbb{P}} = \left(\mathfrak{S}_{\Psi, \Lambda} \int_K F + o(1) \right) \frac{n^d}{(\log n)^t}. \quad (20)$$

The normalized formulation is what is used below; (20) is only a translation. In the auxiliary residue expansion one sums only over lifts on which every displayed form is a unit modulo W_{aux} ; the fixed typed cells modulo W_0 are the residue-compatible cells already shown admissible in the finite-core construction.

The edge-total system for a type $\tau = (a, b, \sigma, r, r', \pi)$ consists of

$$q, \quad q', \quad P_\tau(q, q') = \sigma(bq' - aq). \quad (21)$$

The coefficient vectors $(1, 0), (0, 1), (-\sigma a, \sigma b)$ are pairwise non-proportional. Applying GTZ with weight $g_\tau(q/n, q'/n)$ on the support polygon gives the first moments. Because the typed kernels were defined by dividing the aggregate edge intensity by the one-edge local density, these first moments are exactly the limiting load integrals in (15c)–(15d), up to $o(n/\log n)$.

For the label second moment, two types

$$\tau_i = (a_i, b_i, \sigma_i, r_i, r'_i, \pi)$$

sharing the same label are parametrized on the fixed affine lattice

$$a_i q_i + \sigma_i P \equiv 0 \pmod{b_i}$$

by variables (P, q_1, q_2) , together with the full typed residue constraints

$$P \equiv \pi \pmod{W}, \quad q_i \equiv r_i \pmod{W}, \quad \frac{a_i q_i + \sigma_i P}{b_i} \equiv r'_i \pmod{W}.$$

The last congruence is part of the affine lattice; it is not implied by the divisibility congruence when b_i has factors in W . The prime forms are

$$P, \quad q_1, \quad \frac{a_1 q_1 + \sigma_1 P}{b_1}, \quad q_2, \quad \frac{a_2 q_2 + \sigma_2 P}{b_2}. \quad (22)$$

Restricted to that lattice, these are integer-valued affine-linear forms. The coefficient vectors in (P, q_1, q_2) are

$$(1, 0, 0), (0, 1, 0), (\sigma_1/b_1, a_1/b_1, 0), (0, 0, 1), (\sigma_2/b_2, 0, a_2/b_2),$$

and no two are rational multiples.

For the X -side second moment, two edges sharing $x = 2aq$ necessarily have the same a and the same prime q ; otherwise $a_1 q_1 = a_2 q_2$ is impossible for large n , since a_i are S -smooth while $q_i \notin S$ are primes. The prime forms in variables (q, q'_1, q'_2) are

$$q, \quad q'_1, \quad q'_2, \quad \sigma_1(b_1 q'_1 - a q), \quad \sigma_2(b_2 q'_2 - a q). \quad (23)$$

The coefficient vectors are

$$(1, 0, 0), (0, 1, 0), (0, 0, 1), (-\sigma_1 a, \sigma_1 b_1, 0), (-\sigma_2 a, 0, \sigma_2 b_2),$$

again pairwise non-proportional. The Y -side moment is symmetric, with variables (q', q_1, q_2) and forms

$$q', \quad q_1, \quad q_2, \quad \sigma_1(b q' - a_1 q_1), \quad \sigma_2(b q' - a_2 q_2). \quad (24)$$

These also have pairwise non-proportional coefficient vectors after the same trivial check.

The full affine lattices in (22)–(24) include all fixed congruences modulo $2W$ and the divisibility congruences needed to make the displayed rational forms integral. They are locally admissible. At primes dividing $2W$, the prescribed residue classes make every displayed prime form a unit. At primes outside $2W$, none of the coefficients a_i, b_i vanish, and for each fixed shared variable the remaining representation variables avoid at most two residue classes. Hence there is always an allowed residue tuple.

The only collisions excluded above are lower-dimensional repeated-edge diagonals. Their total weighted contribution is bounded by

$$O\left(\frac{\log^4 n}{n^2} \cdot \frac{n^2}{\log^3 n}\right) = O(\log n) = o(n/\log n). \quad (25)$$

Boundary layers are handled by first replacing every support polygon by a slightly shrunken finite union of rational polytopes, applying GTZ there, and then letting the shrinkage tend to zero.

Bounded piecewise-continuous kernels are treated by L^1 approximation by step functions on rational polytopes. Since the coefficient core and the type set are finite, all such errors remain $o(n/\log n)$ after summing over types and type-pairs.

It remains to spell out why the second moments are the L^2 norms of the limiting loads. In the normalized W_{aux} -tricked formulation, the one-edge density assigned to a typed cell is precisely the density divided out in the definition of g_τ . Therefore the five-form count (22), after disintegration by $t = P/n$ and $\pi = P \bmod W$, has main term

$$\frac{n}{\log n} \sum_{\pi \in \mathcal{B}} \zeta_\pi \int_{1/5}^\beta L_{Z, \tau_1}^{\text{lim}}(t, \pi) L_{Z, \tau_2}^{\text{lim}}(t, \pi) dt + o(n/\log n),$$

and summing over τ_1, τ_2 gives

$$\sum_{P \in Z_n} L_Z(P)^2 = \frac{n}{\log n} \sum_{\pi \in \mathcal{B}} \zeta_\pi \int_{1/5}^\beta (L_Z^{\text{lim}}(t, \pi))^2 dt + o(|Z_n|).$$

Since $L_Z^{\text{lim}} = 1$ and the prime number theorem in the fixed robust classes gives

$$|Z_n| = \frac{n}{\log n} \sum_{\pi \in \mathcal{B}} \zeta_\pi (\beta - 1/5) + o(n/\log n),$$

this proves the label second moment. The corresponding first moment is obtained from (21), hence (17) follows.

The X -side is identical in structure. GTZ applied to (23) gives

$$\sum_{x \in X_n} L_X(x)^2 = \frac{n}{\log n} \sum_{(a,r)} \xi_{a,r} \int (L_X^{\text{lim}}(z, a, r))^2 dz + o(|X_n|).$$

The edge-total estimate (21), now with the bounded test function $L_X^{\text{lim}}(2aQ, a, r)$ inserted, gives the mixed term

$$\sum_{x \in X_n} L_X(x) L_X^{\text{lim}}(x) = \frac{n}{\log n} \sum_{(a,r)} \xi_{a,r} \int (L_X^{\text{lim}}(z, a, r))^2 dz + o(|X_n|),$$

and the fixed-modulus prime number theorem gives the same expression for $\sum_x (L_X^{\text{lim}}(x))^2$. Expanding the square proves (18). The proof of (19) from (24) is symmetric.

If one insists on writing everything in the raw singular-series form (20), the same argument requires checking the finite list of conditional local-factor identities

$$\lambda_{\tau_1, \tau_2}^Z = \frac{\lambda_{\tau_1} \lambda_{\tau_2}}{\zeta_\pi}, \quad \lambda_{\tau_1, \tau_2}^X = \frac{\lambda_{\tau_1} \lambda_{\tau_2}}{\xi_{a,r}}, \quad \lambda_{\tau_1, \tau_2}^Y = \frac{\lambda_{\tau_1} \lambda_{\tau_2}}{\eta_{b,r'}}. \quad (26)$$

These identities say only that, after conditioning on the shared label or side vertex residue cell, the two remaining representation variables are locally independent. For primes $\ell \nmid 2W$, this can be checked directly. The one-edge system has

$$(\ell - 1)(\ell - 2)$$

good pairs $(q, q') \pmod{\ell}$, giving local factor $\ell(\ell - 2)/(\ell - 1)^2$. A label-pair system has, for each nonzero $P \pmod{\ell}$, two independent choices q_i each avoiding 0 and one further nonzero class, hence $(\ell - 1)(\ell - 2)^2$ good triples and local factor $[\ell(\ell - 2)/(\ell - 1)^2]^2$. The X - and Y -pair systems are identical after conditioning on the shared side vertex. The finitely many primes dividing $2W$ are fixed residue-cell checks built into the affine lattices. Thus (26) holds, and the normalized formulation above is simply a concise way to organize the same finite verification. $\square$

5 Fractional preprocessing and Kahn rounding

The following deterministic lemma converts Proposition 5 into a large fractional matching.

Lemma 6 (Fractional preprocessing). *There are subweights $0 \leq t_e \leq w_e$ such that*

$$\sum_{e \ni v} t_e \leq 1 \quad \text{for every vertex } v, \quad \sum_e t_e = (1 - o(1))|Z_n|, \quad (27)$$

$$\max_e t_e = o(1), \quad a(t) := \max_{u \neq v} \sum_{e \ni \{u,v\}} t_e = o(1). \quad (28)$$

Proof. Normalize labels. For $P \in Z_n$, set

$$c_P = \min(1, L_Z(P)^{-1})$$

with $c_P = 1$ if $L_Z(P) = 0$, and put $w'_e = c_P w_e$ for edges with label P . Then

$$\sum_e w'_e = \sum_{P \in Z_n} \min(L_Z(P), 1) = |Z_n| - o(|Z_n|), \quad (29)$$

because $1 - \min(u, 1) \leq |1 - u|$ and Cauchy–Schwarz applies to (17). Moreover every normalized label load is at most 1.

Since $w'_e \leq w_e$, side loads only decrease. Let $B_X = \{x : L'_X(x) > 1\}$. Because $L_X^{\lim} \leq 1 - 2\gamma$, (18) implies not only $|B_X| = o(|X_n|)$ but also

$$\sum_{x \in B_X} L'_X(x) = o(|Z_n|). \quad (30)$$

Indeed $B_X \subset \{x : L_X(x) > 1\}$, and on this latter set $L_X(x) - L_X^{\lim}(x) > 2\gamma$, so (18) gives $|B_X| = o(|X_n|)$. Also, writing $\Delta_X(x) = L_X(x) - L_X^{\lim}(x)$ and using $L'_X \leq L_X$,

$$\sum_{x \in B_X} L'_X(x) \leq |B_X| + \sum_{x \in B_X} |\Delta_X(x)| \leq |B_X| + |B_X|^{1/2} \left(\sum_{x \in X_n} \Delta_X(x)^2 \right)^{1/2} = o(|Z_n|),$$

since $|X_n| \asymp |Z_n|$. The same argument gives

$$\sum_{y \in B_Y} L'_Y(y) = o(|Z_n|). \quad (31)$$

Delete all edges incident to $B_X \cup B_Y$ and call the remaining weights t_e . Then all vertex loads are at most 1, and (29)–(31) give $\sum_e t_e = (1 - o(1))|Z_n|$. Also

$$t_e \leq w_e \ll \frac{\log^2 n}{n} = o(1).$$

Finally the actual hypergraph has pair codegree at most 2: a pair (x, y) determines $P = |x - y|/2$, and a pair (x, P) or (y, P) has at most the two opposite-side possibilities $x \pm 2P$ or $y \pm 2P$. Hence

$$a(t) \leq 2 \max_e t_e = o(1).$$

□

We now use Kahn's theorem. In the notation of [4], for a hypergraph $\mathcal{H}$ and a fractional matching t , set

$$a(t) = \max_{x \neq y} \sum_{A \ni x, y} t(A). \quad (32)$$

Kahn's Theorem 1.5 says: for fixed k and l , if $\mathcal{H}$ is k -bounded, t is a fractional matching, and $c_1, \dots, c_l : \mathcal{H} \rightarrow \mathbb{R}_+$ satisfy

$$\sum_{A \in \mathcal{H}} c_i(A)^2 t(A) = o \left(\left(\sum_{A \in \mathcal{H}} c_i(A) t(A) \right)^2 \right), \quad (33)$$

then as $a(t) \rightarrow 0$ there is a matching M with

$$\sum_{A \in M} c_i(A) \sim \sum_{A \in \mathcal{H}} c_i(A) t(A)$$

for every i .

Apply this with $k = 3$, $l = 1$, and $c_1(e) \equiv 1$. Then

$$\sum_e c_1(e)^2 t_e = \sum_e t_e = o \left(\left(\sum_e t_e \right)^2 \right)$$

because $\sum_e t_e \asymp |Z_n| \rightarrow \infty$. Thus there is a genuine matching M_n with

$$|M_n| = (1 - o(1)) |Z_n|. \quad (34)$$

6 Cleanup and completion

For each matched edge (x, y, P) choose

$$a_P \equiv x \pmod{P}.$$

Since $|x - y| = 2P$, this also hits y . The residue is nonzero: if $P > n/5$ divided an even residual target $x \leq n$, then $x = 2P$ or $x = 4P$, but these are not residual because robustness gives two S -hits.

Let

$$N = |A_S(n)| = (1 + o(1)) \frac{n}{\log n}.$$

By (9) and (34),

$$|M_n| = ((\beta - 1/5)\delta_S + o(1)) N. \quad (35)$$

The full robust reservoir satisfies

$$|\mathcal{R}(n)| = \left(\frac{4}{5} \delta_S + o(1) \right) N. \quad (36)$$

After the pair stage, the number of unused robust primes is $|\mathcal{R}(n)| - |M_n|$. The number of residual tokens still uncovered, including the coefficient tails and all exceptional tokens, is at most $N - 2|M_n| + o(N)$. It is enough that

$$|\mathcal{R}(n)| - |M_n| \geq N - 2|M_n| + o(N),$$

or

$$|M_n| \geq N - |\mathcal{R}(n)| + o(N). \quad (37)$$

This follows from the strict inequality

$$(\beta - 1/5)\delta_S > 1 - \frac{4}{5}\delta_S,$$

which is exactly (7).

Assign each remaining residual token injectively to an unused robust prime P and choose

$$a_P \equiv x \pmod{P}.$$

If one integer has two residual tokens, use two distinct unused robust primes. The residue is nonzero by the same $2P, 4P$ argument; odd exceptional tokens cannot be P or $3P$, because P is already covered by parity plus an S -hit and $3P$ is covered by parity plus the zero class modulo 3. The robust side-debt lemma ensures that switching these primes creates no new unresolved debt.

All residual tokens are defined after the fixed choices for 2, 3, and S . The matching and singleton assignments discharge these tokens injectively. The only new debts created by the cleanup primes are multiples of those cleanup primes, and every such multiple is handled by the robust side-debt lemma. All other primes remain in their zero classes, 2 remains in the odd class, 3 remains in the zero class, and the primes in S remain in their fixed nonzero classes. The exact debt inequality (1) is satisfied for every $m \leq n$, so every $m \in [1, n]$ lies in at least two chosen residue classes.

Theorem 7. *For all sufficiently large n , there are residue classes $a_p \pmod{p}$ for all primes $p \leq n$ such that every integer $m \in [1, n]$ satisfies*

$$\#\{p \leq n : m \equiv a_p \pmod{p}\} \geq 2.$$

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